

Semisupervised Laplace-Regularized Multimodality Metric Learning

Jianqing Liang^{1b}, Pengfei Zhu^{1b}, *Member, IEEE*, Chuangyin Dang^{2b}, *Senior Member, IEEE*,
and Qinghua Hu^{1b}, *Senior Member, IEEE*

Abstract—Distance metric learning, which aims at learning an appropriate metric from data automatically, plays a crucial role in the fields of pattern recognition and information retrieval. A tremendous amount of work has been devoted to metric learning in recent years, but much of the work is basically designed for training a linear and global metric with labeled samples. When data are represented with multimodal and high-dimensional features and only limited supervision information is available, these approaches are inevitably confronted with a series of critical problems: 1) naive concatenation of feature vectors can cause the curse of dimensionality in learning metrics and 2) ignorance of utilizing massive unlabeled data may lead to overfitting. To mitigate this deficiency, we develop a semisupervised Laplace-regularized multimodal metric-learning method in this work, which explores a joint formulation of multiple metrics as well as weights for learning appropriate distances: 1) it learns a global optimal distance metric on each feature space and 2) it searches the optimal combination weights of multiple features. Experimental results demonstrate both the effectiveness and efficiency of our method on retrieval and classification tasks.

Index Terms—Laplace regularized, metric learning, multimodal, semisupervised.

I. INTRODUCTION

THE EXTENSIVE popularity of social networks and websites has stimulated the explosive growth of multimedia objects, which are described with multiple modalities of features. For example, a piece of news can be represented with texts, images, and videos [1]; videos include audio and visual signals; the musical data may be associated with semantic features (lyrics and themes), acoustic features (rhythm),

and social features (artist biographies) [2]. Each modality is characterized by distinct intrinsic properties of an object.

It is crucial to define an appropriate distance metric in a wide range of applications, including information retrieval [3]–[6], bioinformatics [7], and computer vision [8]–[12]. Related studies show that proper metrics can make significant improvement in multimedia clustering [13]–[16] and k -nearest neighbor (k NN) classification [11], [15], [17]–[19]. Unfortunately, as a commonly used metric, the Euclidean distance may not be always optimal for mining patterns in complex sceneries due to the ignorance of data structures and the semantic gap between low-level features and high-level human comprehension [20], [21]. This also explains why the Euclidean distance-based methods, like k -means, may fail in such cases [22].

To address the issues above, there has been considerable research on learning task-specific metrics in the past few years [13], [17], [23]–[28]. Xing *et al.* [13] first formulated metric learning as a constrained convex optimization problem, where a linear global distance metric is obtained by pushing the dissimilar pairs far from each other. This pioneering work has inspired an increasing amount of work on learning global metrics, including nearest-neighbors-driven methods [17], [23], [24]; information-theoretic methods [25], [26]; and online learning approaches [27], [28].

Although excellent performance has been achieved in a variety of practical applications, this kind of global metrics suffers from several limitations in measuring similarity for objects described with multiple modalities [29]–[32]. The naive concatenation approach may lead to a high-dimensional feature space, thus causing the curse of dimensionality [20], [22]. Moreover, due to the limited computing and memory resources, the applicability of global metrics is weakened in large-scale and high-dimensional problems. While a surge of efforts has been put on dimensionality reduction [33]–[36], these single-modal metrics neglect capturing the complementarity and consensus properties among different modalities.

In order to complete related tasks, such as multimodal high-dimensional classification, clustering and retrieval in machine learning, and computer vision, it is highly desired to explore effective and efficient metrics for multimodal data. Under such circumstances, multiple kernel techniques, which project data to high-dimensional Hilbert spaces with non-linear kernel functions, have been investigated over recent decades [2], [37], [38]. McFee and Lanckriet [2] first proposed integrating multiple modalities into a unified feature space through an optimal ensemble of kernel transformations.

Manuscript received August 16, 2019; revised January 31, 2020; accepted August 3, 2020. Date of publication October 7, 2020; date of current version May 19, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62006147, Grant 61925602, Grant 61732011, Grant 61976184, and Grant 61876103. This article was recommended by Associate Editor Y. Jin. (*Corresponding authors: Pengfei Zhu; Qinghua Hu.*)

Jianqing Liang is with the School of Computer and Information Technology, Shanxi University, Taiyuan 030006, China, and also with the College of Intelligence and Computing, Tianjin University, Tianjin 300072, China (e-mail: liangjianqing@tju.edu.cn).

Pengfei Zhu and Qinghua Hu are with the College of Intelligence and Computing, Tianjin University, Tianjin 300072, China (e-mail: zhupengfei@tju.edu.cn; huqinghua@tju.edu.cn).

Chuangyin Dang is with the Department of Systems Engineering and Engineering Management, City University of Hong Kong, Hong Kong (e-mail: mecdang@cityu.edu.hk).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCYB.2020.3022277>.

Digital Object Identifier 10.1109/TCYB.2020.3022277

However, due to extensive computation, it is not scalable to large-scale applications. To improve efficiency, online learning techniques have been introduced in recent years [3], [39]. In [39], an online multiple kernel similarity (OMKS) learning scheme was proposed to learn a proximity function. In [3], ensemble learning and active sample selection were introduced to further improve the performance of OMKs. Despite their obvious advantage over global methods, they may fall short in the scalability for large-scale problems [21]. In addition, the majority of current multimodal metric-learning methods ignores the usage of unlabeled data, which leads to the problem of overfitting when only a few labeled data are provided.

Since the acquisition of labeled data requires human resources, there are still massive unlabeled data to be explored in practical applications [40]. Hence, learning an effective metric with a small portion of supervision information has become an active research topic over the past decade [11], [41], [42]. Liu *et al.* [41] proposed a constrained metric-learning algorithm to integrate synergistic information from both the labeled and unlabeled data. It can be applied to broad applications, such as classification, image retrieval, and k -means clustering. Wang [42] learned a distance metric by maximally unfolding the data in the projection space and meanwhile satisfying pairwise constraints. Although a great deal of effort has been invested on semisupervised distance metric learning, there are still several problems that remain to be solved. First, quite a number of the methods plunge into a local optimum. Second, most of the methods are not applicable to high-dimensional problems due to the lack of a regularizer. Last but not least, these linear global metrics are not competent in capturing the underlying structure in multimodal data. These shortcomings appeal for a more effective strategy to learn a metric on multimodal high-dimensional data in a semisupervised manner.

In this article, we propose a semisupervised Laplace-regularized multimodal metric-learning (SLRMML) framework to learn metrics for multimodal high-dimensional data. First, we consider the single-modal problem and investigate the single-modal semisupervised Laplace-regularized metric-learning (SLRML) method. SLRML seeks for a lower-dimensional feature space which maximizes the interclass scatter and minimizes the intraclass scatter. To preserve the original manifold structure, we exploit both labels and data in the construction of a semisupervised Laplace graph regularizer. A global solution is obtained from an eigenvalue decomposition strategy. Then, we extend the method to multimodal tasks and derive the SLRMML method. The corresponding weight associated with each modality is learned to distinguish its importance via a joint optimization objective. Compared with the state-of-the-art metric-learning methods based on multiple kernel techniques, the SLRMML is more effective and efficient in learning metrics for high-dimensional multimodal data. Our contributions are summarized as follows.

- 1) We propose a general framework of semisupervised Laplace-regularized metric learning, which is suitable for high-dimensional data. Specifically, our approach can reduce the dimensionality and make full use of massive

unlabeled data. To the best of our knowledge, this is the first attempt of applying semisupervised Laplace to graph regularizer in metric learning.

- 2) We extend our method to multimodal data by learning the feature mappings and combination weights for multiple modalities in a joint framework. For each modality, an optimal lower-dimensional feature space is obtained through eigenvalue decomposition. Meanwhile, its dimensionality can be flexibly automatically determined.
- 3) Experimental results on both retrieval and classification tasks validate the advantages of the proposed method in terms of accuracy and scalability.

The remainder of this article is arranged as follows. Section II gives a brief introduction of metric learning. Section III describes and addresses the semisupervised metric-learning problem. An extension of multimodal data is presented in Section IV. Section V validates the effectiveness and efficiency of SLRML and SLRMML through extensive experiments. The conclusion and outlook are summarized in Section VI.

II. DISTANCE METRIC LEARNING

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$ be a matrix of N data points, where $\mathbf{x}_i \in \mathbb{R}^D$ is a D -dimensional feature vector. For labeled data, two sets of pairwise constraints, including positive pairs and negative pairs, respectively, are defined as

$$\begin{aligned} \mathcal{S} &:= \{(\mathbf{x}_i, \mathbf{x}_j) | \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are in the same class}\} \\ \mathcal{D} &:= \{(\mathbf{x}_i, \mathbf{x}_j) | \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are in different classes}\}. \end{aligned}$$

Our goal is to learn a Mahalanobis distance

$$d_{\mathbf{M}}(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{(\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{M} (\mathbf{x}_i - \mathbf{x}_j)}$$

where $\mathbf{M}$ is a $D \times D$ real and symmetric square matrix to be determined through learning. According to the properties of metric, $d_{\mathbf{M}}$ must satisfy non-negativity, identity, symmetry, as well as triangle inequality [20].

The distance metric-learning problem basically aims to learn an appropriate metric $\mathbf{M}$ from a data set $\mathcal{X}$ that best agrees with a set of positive pairs $\mathcal{S}$ and a set of negative pairs $\mathcal{D}$ to approximate the underlying data structure, which can be typically formulated as the following optimization framework:

$$\min_{\mathbf{M} \succeq 0} \ell(\mathbf{M}, \mathcal{S}, \mathcal{D}) + \lambda R(\mathbf{M})$$

where $\ell(\mathbf{M}, \mathcal{S}, \mathcal{D})$ is a loss function that generates penalty when the constraints are violated, $R(\mathbf{M})$ is a regularizer on $\mathbf{M}$, and λ is a non-negative tradeoff parameter.

III. SEMISUPERVISED LAPLACE-REGULARIZED METRIC LEARNING

In this section, we describe our proposed SLRML method in detail.

A. Maximum Margin Criterion

First, we introduce the concept of constraint margin [42].

Definition 1 (Constraint Margin): Suppose there are two sets of constraints $\mathcal{S}$ and $\mathcal{D}$ generated from $\mathbf{X}$, we define constraint margin as

$$\delta = \sum_{(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{D}} \frac{d_M^2(\mathbf{x}_i, \mathbf{x}_j)}{N_D} - \sum_{(\mathbf{x}_k, \mathbf{x}_l) \in \mathcal{S}} \frac{d_M^2(\mathbf{x}_k, \mathbf{x}_l)}{N_S} \quad (1)$$

where N_D and N_S are the numbers of negative pairs and positive pairs, respectively.

Intuitively, the first term in (1) represents the mean of squared distances between different classes while the second term represents the mean of squared distances within the same class. That is, δ stands for the gap of the between-class scatterness and the within-class scatterness, which should be maximized in search of an appropriate metric. It can be rewritten as $\delta = \sum_i \delta_i$ with

$$\delta_i = \sum_{(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{D}} \frac{d_M^2(\mathbf{x}_i, \mathbf{x}_j)}{N_D} - \sum_{(\mathbf{x}_i, \mathbf{x}_k) \in \mathcal{S}} \frac{d_M^2(\mathbf{x}_i, \mathbf{x}_k)}{N_S}$$

where δ_i is the constraint margin of the data point $\mathbf{x}_i$.

In practical applications, there exists a large amount of high-dimensional data, such as image and text, which leads to high computational cost and storage requirements. To deal with such datasets, it is highly desirable to consider metric learning and dimensionality reduction from a unified perspective [22].

As $\mathbf{M}$ is a symmetric positive semidefinite matrix, we can decompose $\mathbf{M} = \mathbf{U}\mathbf{U}^T$, where $\mathbf{U} \in \mathbb{R}^{D \times d}$ is the transformation matrix. With this decomposition, one can rewrite $d_M^2(\mathbf{x}_i, \mathbf{x}_j)$ as

$$\begin{aligned} d_M^2(\mathbf{x}_i, \mathbf{x}_j) &= (\mathbf{x}_i - \mathbf{x}_j)^T \mathbf{U} \mathbf{U}^T (\mathbf{x}_i - \mathbf{x}_j) \\ &= (\mathbf{U}^T (\mathbf{x}_i - \mathbf{x}_j))^T (\mathbf{U}^T (\mathbf{x}_i - \mathbf{x}_j)) \\ &= (\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j)^T (\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j) \end{aligned}$$

where $\tilde{\mathbf{x}}_i = \mathbf{U}^T \mathbf{x}_i$. Therefore, the Mahalanobis metric-learning method can be regarded as the Euclidean distance in the projected feature space. As $d < D$, a lower-dimensional feature space is generated for dimensionality reduction.

Let

$$\mathbf{D} = \sum_{(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{D}} \frac{(\mathbf{x}_i - \mathbf{x}_j)(\mathbf{x}_i - \mathbf{x}_j)^T}{N_D} \quad (2)$$

$$\mathbf{S} = \sum_{(\mathbf{x}_k, \mathbf{x}_l) \in \mathcal{S}} \frac{(\mathbf{x}_k - \mathbf{x}_l)(\mathbf{x}_k - \mathbf{x}_l)^T}{N_S} \quad (3)$$

be the between-class scatterness and within-class scatterness matrices, respectively. With $\mathbf{D}$ and $\mathbf{S}$, δ can be represented as

$$\begin{aligned} \delta &= \text{tr} \left(\mathbf{U}^T \sum_{(\mathbf{x}_i, \mathbf{x}_j) \in \mathcal{D}} \frac{(\mathbf{x}_i - \mathbf{x}_j)(\mathbf{x}_i - \mathbf{x}_j)^T}{N_D} \mathbf{U} \right. \\ &\quad \left. - \mathbf{U}^T \sum_{(\mathbf{x}_k, \mathbf{x}_l) \in \mathcal{S}} \frac{(\mathbf{x}_k - \mathbf{x}_l)(\mathbf{x}_k - \mathbf{x}_l)^T}{N_S} \mathbf{U} \right) \\ &= \text{tr}(\mathbf{U}^T (\mathbf{D} - \mathbf{S}) \mathbf{U}). \end{aligned}$$

Our purpose is to find a transformation matrix $\mathbf{U}$ that maximizes δ .

B. Semisupervised Laplace Graph Regularizer

To preserve the supervision information as well as geometry structure in the original space, we introduce a graph regularizer by taking advantage of both labeled data and unlabeled data.

Given a matrix of N data points $\mathbf{X}$, we may compute a similarity matrix based on distances

$$W_{ij}^u = \begin{cases} \frac{1}{Nk}, & \mathbf{x}_i \in \mathcal{N}_k(\mathbf{x}_j) \text{ or } \mathbf{x}_j \in \mathcal{N}_k(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases}$$

where $\mathcal{N}_k(\mathbf{x}_i)$ denotes a k NN set of $\mathbf{x}_i$.

Similarly, an adjacency matrix based on the supervision information is set as

$$W_{ij}^l = \begin{cases} \frac{1}{2N_S}, & i \neq j, \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are in a same class} \\ 0, & \text{otherwise.} \end{cases}$$

That is, for all the unlabeled data pairs, the corresponding elements of matrix $\mathbf{W}^l$ are zero.

Then, the unsupervised and supervised Laplace matrix can be constructed as

$$\mathbf{L}^h = \mathbf{D}^h - \mathbf{W}^h \quad (4)$$

where $\mathbf{D}_{ii}^h = \sum_j W_{ij}^h$, $h = u, l$.

According to the linearly additive property of Laplace matrices, we obtain a semisupervised Laplace matrix

$$\mathbf{L}^s = \mathbf{L}^u + \mathbf{L}^l. \quad (5)$$

Here, we treat the supervised and unsupervised Laplace matrices equally for the lack of prior knowledge.

In fact, take (4) into (5), we have

$$\begin{aligned} \mathbf{L}^s &= \mathbf{D}^u - \mathbf{W}^u + \mathbf{D}^l - \mathbf{W}^l \\ &= \mathbf{D}^s - \mathbf{W}^s \end{aligned}$$

where $\mathbf{W}^s = \mathbf{W}^u + \mathbf{W}^l$ and $\mathbf{D}^s = \mathbf{D}^u + \mathbf{D}^l$.

Finally, we obtain the graph regularizer

$$\sum_{i,j} \|\mathbf{U}^T \mathbf{x}_i - \mathbf{U}^T \mathbf{x}_j\|^2 W_{ij}^s = \text{tr}(\mathbf{U}^T \mathbf{X} \mathbf{L}^s \mathbf{X}^T \mathbf{U})$$

which should be minimized in order to preserve the consistency of the manifold structure and the supervision information.

The element in matrix $\mathbf{W}^s$ is composed of 0, $(1/Nk)$, $(1/2N_S)$, and $(1/Nk) + (1/2N_S)$. For labeled-labeled pairs, those belong to the same class and lie in the neighborhood are regarded as the most relevant with similarity $(1/Nk) + (1/2N_S)$. Conversely, pairs that are in different classes and not neighbors are regarded as dissimilar with similarity 0. For pairs of which the labels and distance are inconsistent, that is, non-neighbors in the same class or neighbors in different classes, there is a compromise and the similarity is set as $(1/2N_S)$ and $(1/Nk)$, respectively. The cases of the remaining three types, including unlabeled-unlabeled pairs, labeled-unlabeled pairs, and unlabeled-labeled pairs, are much simpler. The similarity is set as $(1/Nk)$ if the pairs are neighbors and 0 otherwise. The semisupervised Laplace graph regularizer is constructed with the similarity matrix $\mathbf{W}^s$. Fig. 1 gives the significance of the regularizer.

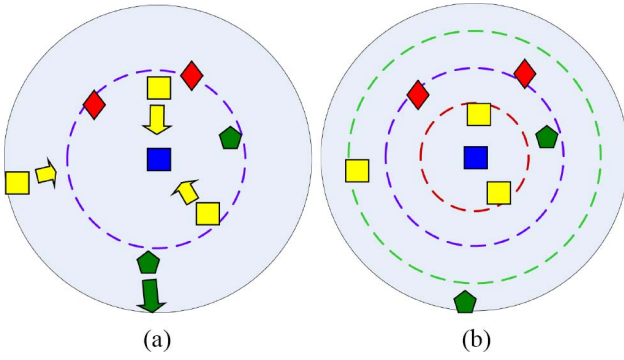


Fig. 1. Intuitive illustration of semisupervised Laplace graph regularizer in terms of labeled-labeled pairs. The dashed lines of three colors, that is, red, blue, and green, represent the similarities (i.e., element in matrix $\mathbf{W}^s$) $(1/Nk) + (1/2N_S)$, $(1/Nk)$, and $(1/2N_S)$, respectively. The blue square stands for a labeled data point $\mathbf{x}_i$. The yellow squares represent the points that belong to the same class with $\mathbf{x}_i$, while the red diamonds and green pentagons are those which are in different classes (a) gives the data distribution of labeled-labeled pairs with unsupervised Laplace graph regularizer $\text{tr}(\mathbf{U}^T \mathbf{X} \mathbf{L}^u \mathbf{X}^T \mathbf{U})$, which preserves the manifold structure in the Euclidean space and (b) gives the data distribution of labeled-labeled pairs with semisupervised Laplace graph regularizer $\text{tr}(\mathbf{U}^T \mathbf{X} \mathbf{L}^s \mathbf{X}^T \mathbf{U})$.

C. Objective and Solution

With the aforementioned term and regularizer, our SLRML method can be formulated as

$$\begin{aligned} \max_{\mathbf{U}} \quad & \text{tr}(\mathbf{U}^T (\mathbf{D} - \mathbf{S}) \mathbf{U}) - \alpha \left(\text{tr}(\mathbf{U}^T \mathbf{X} \mathbf{L}^s \mathbf{X}^T \mathbf{U}) + \text{tr}(\mathbf{U}^T \mathbf{U}) \right) \\ \text{s.t.} \quad & \mathbf{U}^T \mathbf{U} = \mathbf{I}. \end{aligned} \quad (6)$$

The regularizer $\text{tr}(\mathbf{U}^T \mathbf{U})$ is introduced to control the complexity of the model. α is a tradeoff parameter that balances the importance between the margin and regularizer. The equality constraint is used to limit the scale of the transformation matrix. The dimensionality of the embedding feature space is determined by the following theorem [43].

Theorem 1 (Ky Fan): Let $\mathbf{H}$ be a symmetric matrix with its eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$$

and corresponding eigenvectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_d]$. Then, we have

$$\lambda_1 + \lambda_2 + \dots + \lambda_k = \max_{\mathbf{Q}^T \mathbf{Q} = \mathbf{I}} \text{tr}(\mathbf{Q}^T \mathbf{H} \mathbf{Q}).$$

Furthermore, the optimal $\mathbf{Q}^*$ is $\mathbf{Q}^* = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k]$.

In fact, the objective in (6) can be reformulated as

$$\begin{aligned} \max_{\mathbf{U}} \quad & \text{tr} \left[\mathbf{U}^T (\mathbf{D} - \mathbf{S} - \alpha (\mathbf{X} \mathbf{L}^s \mathbf{X}^T + \mathbf{I})) \mathbf{U} \right] \\ \text{s.t.} \quad & \mathbf{U}^T \mathbf{U} = \mathbf{I}. \end{aligned} \quad (7)$$

As $\mathbf{D}$, $\mathbf{S}$, $\mathbf{X} \mathbf{L}^s \mathbf{X}^T$, and $\mathbf{I}$ are all symmetric matrices, we may conclude that the optimal dimensionality d should be the number of positive eigenvalues, and the transformation matrix is composed of the corresponding eigenvectors of $\mathbf{D} - \mathbf{S} - \alpha (\mathbf{X} \mathbf{L}^s \mathbf{X}^T + \mathbf{I})$. The complete procedure of the method is summarized in Algorithm 1.

Algorithm 1 SLRML

Input: Training data $\mathbf{X}$, constraint sets $\mathcal{S}$, $\mathcal{D}$, parameter α .

- 1: Compute the between-class scatterness matrix $\mathbf{D}$, within-class scatterness matrix $\mathbf{S}$ and semi-supervised Laplace matrix $\mathbf{L}$ with Eq.(2), Eq.(3), Eq.(4) and Eq.(5);
- 2: Perform eigenvalue decomposition on $\mathbf{D} - \mathbf{S} - \alpha (\mathbf{X} \mathbf{L}^s \mathbf{X}^T + \mathbf{I})$;
- 3: Search the d non-negative eigenvalues and then select the corresponding eigenvectors to form $\mathbf{U}$;

Output: $\mathbf{U}$.

IV. EXTENSION TO MULTIMODAL DATA

In the previous section, we have presented the SLRML method, which learns a linear global Mahalanobis metric by taking advantage of maximum margin criterion and semisupervised Laplace graph regularizer. The SLRML is able to tackle the high-dimensional problem with matrix decomposition. Nevertheless, when handling multimodal tasks in the fields of biometric recognition, face recognition, and information retrieval [7], [10], [21], [44]–[46], it neglects the consensus and complementarity properties between different modalities. To address this issue, we extend the SLRML to multimodality data by introducing combination weights in this section.

A. Objective

Suppose each data point $\mathbf{x}_i \in \mathbf{X}$ is represented with m modalities $\mathbf{x}_i^1, \mathbf{x}_i^2, \dots, \mathbf{x}_i^m$. We aim to learn a weighted Mahalanobis distance

$$d_{\{\mathbf{M}_p, w_p\}_{p=1}^m}^2(\mathbf{x}_i, \mathbf{x}_j) = \sum_{p=1}^m w_p (\mathbf{x}_i^p - \mathbf{x}_j^p)^T \mathbf{M}_p (\mathbf{x}_i^p - \mathbf{x}_j^p)$$

where w_p is the weight of the p th modality in learning metrics, and $\mathbf{M}_p$ is the corresponding $D_p \times D_p$ positive semidefinite matrix to be solved. For the p th modality, supervision information is given in the form of positive and negative pairs

$$\begin{aligned} \mathcal{S}_p &:= \left\{ (\mathbf{x}_i^p, \mathbf{x}_j^p) \mid \mathbf{x}_i^p \text{ and } \mathbf{x}_j^p \text{ are in the same class} \right\} \\ \mathcal{D}_p &:= \left\{ (\mathbf{x}_i^p, \mathbf{x}_j^p) \mid \mathbf{x}_i^p \text{ and } \mathbf{x}_j^p \text{ are in different classes} \right\}. \end{aligned}$$

As an extension of the objective (7), an SLRMML method can be stated as

$$\begin{aligned} \max_{\{\mathbf{U}_p, w_p\}_{p=1}^m} \quad & \sum_{p=1}^m w_p^r \text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right] \\ \text{s.t.} \quad & \sum_{p=1}^m w_p = 1, w_p \geq 0, \mathbf{U}_p^T \mathbf{U}_p = \mathbf{I}_p, p = 1, 2, \dots, m \end{aligned} \quad (8)$$

where $\mathbf{U}_p$ is the low-rank projection to be learned for the p th modality and satisfies $\mathbf{M}_p = \mathbf{U}_p \mathbf{U}_p^T$. $\mathbf{X}_p$ is the data matrix on the p th modality and $\mathbf{L}_p^s$ is the corresponding semisupervised Laplace matrix, $\mathbf{I}_p$ is a $D_p \times D_p$ identity matrix, and $\mathbf{D}_p$ and $\mathbf{S}_p$ are the interclass scatter and intraclass scatter of the p th

modality, respectively

$$\mathbf{D}_p = \sum_{(\mathbf{x}_i^p, \mathbf{x}_j^p) \in \mathcal{D}_p} \frac{(\mathbf{x}_i^p - \mathbf{x}_j^p)(\mathbf{x}_i^p - \mathbf{x}_j^p)^T}{N_{\mathcal{D}_p}} \quad (9)$$

$$\mathbf{S}_p = \sum_{(\mathbf{x}_k^p, \mathbf{x}_l^p) \in \mathcal{S}_p} \frac{(\mathbf{x}_k^p - \mathbf{x}_l^p)(\mathbf{x}_k^p - \mathbf{x}_l^p)^T}{N_{\mathcal{S}_p}}. \quad (10)$$

In (8), α is the tradeoff parameter to balance the weights between the importance of margin and regularizers. As for the constraints, w_p is required to be non-negative to ensure the non-negativity of the metric. Furthermore, the sum of w_p equals a constant since each modality is expected to make its own contribution.

B. Solution

Since it is clear that both the transformation matrices and weights are significant in learning a proper metric for multimodality data, SLRMML aims to learn $\mathbf{U}_p$ and w_p simultaneously. To achieve the joint learning of $\mathbf{U}_p$ and w_p , we utilize an alternating procedure to solve the optimization problem (8). The procedure can be divided into two steps.

First, we rewrite the original objective function as the following formulation for fixed w_p :

$$\begin{aligned} \max_{\{\mathbf{U}_p\}_{p=1}^m} \quad & \sum_{p=1}^m w_p^r \text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right] \\ \text{s.t.} \quad & \mathbf{U}_p^T \mathbf{U}_p = \mathbf{I}_p, p = 1, 2, \dots, m. \end{aligned} \quad (11)$$

Due to the independency between $\mathbf{U}_p$'s, (11) is equivalent to a series of problems for $p = 1, 2, \dots, m$

$$\begin{aligned} \max_{\mathbf{U}_p} \quad & w_p^r \text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right] \\ \text{s.t.} \quad & \mathbf{U}_p^T \mathbf{U}_p = \mathbf{I}_p. \end{aligned}$$

According to Theorem 1, the optimal $\mathbf{U}_p$ is formed with the eigenvectors of $w_p^r (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p))$ corresponding to its positive eigenvalues. The following theorem may give us some clues [47].

Theorem 2: Let $p(t)$ be a given polynomial of degree k . If λ, x is an eigenvalue–eigenvector pair of $\mathbf{A} \in \mathbf{M}_n$, then $p(\lambda), x$ is an eigenvalue–eigenvector pair of $p(\mathbf{A})$. Conversely, if $k \geq 1$ and if μ is an eigenvalue of $p(\mathbf{A})$, then there is some eigenvalue λ of $\mathbf{A}$ such that $\mu = p(\lambda)$.

According to the above theorem, for any constant c and matrix $\mathbf{A}$, $c\mathbf{A}$ and $\mathbf{A}$ have the same eigenvector while the eigenvalue of the former is c times of the latter. That is, we can obtain the optimal $\mathbf{U}_p$ by computing the eigenvectors corresponding to the positive eigenvalues of $\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)$, which has the same eigenvectors with $w_p^r (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p))$. In fact, the theorem and analysis guarantee the stability of our approach, that is, no matter what the initialization of $\mathbf{w}$ is, the learned optimal transformation matrix $\mathbf{U}_p$ for each modality maintains unchanged and is composed of only the eigenvectors corresponding to the positive eigenvalues of $\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)$. In view of this, we ignore the initialization step of $\mathbf{w}$ in our method.

Algorithm 2 SLRMML

Input: Training data with m modalities $\{\mathbf{X}_p\}_{p=1}^m$, constraint sets $\{\mathcal{S}_p, \mathcal{D}_p\}_{p=1}^m$, parameter α, r .

- 1: **for** $p = 1$ to m **do**
- 2: Compute the between-class scatterness matrix $\mathbf{D}_p$ and within-class scatterness matrix $\mathbf{S}_p$ with Eq.(9), Eq.(10) and semi-supervised Laplace matrix $\mathbf{L}_p$;
- 3: Perform eigenvalue decomposition on $\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)$;
- 4: Search for the d_p non-negative eigenvalues and then select the corresponding eigenvectors to form $\mathbf{U}_p$;
- 5: **end for**
- 6: Obtain $\mathbf{w}$ by solving Eq.(15);

Output: $\{\mathbf{U}_p\}_{p=1}^m, \{\mathbf{w}_p\}_{p=1}^m$.

Second, we obtain w_p with the learned $\mathbf{U}_p$. We construct a Lagrange function

$$\begin{aligned} L(\mathbf{w}, \eta) = & \sum_{p=1}^m w_p^r \text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right] \\ & - \eta \left(\sum_{p=1}^m w_p - 1 \right). \end{aligned} \quad (12)$$

Then, we let $([\partial L(\mathbf{w}, \eta)]/\partial \eta) = 0$ and $([\partial L(\mathbf{w}, \eta)]/\partial w_p) = 0$, and have

$$\sum_{p=1}^m w_p - 1 = 0 \quad (13)$$

$$r w_p^{r-1} \text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right] - \eta = 0. \quad (14)$$

Combining (13) and (14), we have

$$w_p = \frac{\left(\frac{1}{\text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right]} \right)^{1/(r-1)}}{\sum_{p=1}^m \left(\frac{1}{\text{tr} \left[\mathbf{U}_p^T (\mathbf{D}_p - \mathbf{S}_p - \alpha (\mathbf{X}_p \mathbf{L}_p^s \mathbf{X}_p^T + \mathbf{I}_p)) \mathbf{U}_p \right]} \right)^{1/(r-1)}}. \quad (15)$$

The entire procedure of SLRMML is given in Algorithm 2.

Theorem 3: The objective function (8) has a global optimal solution, which can be derived from the above algorithm.

The proof is given in the Appendix.

V. EXPERIMENTS

In this section, extensive experiments are performed to validate the effectiveness and efficiency of our approach. First, we give a description of datasets, descriptors, experimental settings, and evaluation criteria. Then, we introduce the baseline methods and parameter setting. Finally, we investigate the performance of our proposed SLRML and SLRMML methods on both single modality and multiple modalities tasks for retrieval and classification.

A. Datasets and Descriptors

The experiments are conducted on four single-modal datasets and three multimodal datasets. We choose ten classes and select 300 images for each class from *Nuswide* [48] to form a subset *Nuswide 3k*, where each image is associated with more than one tag. The ten classes are birds, boats, flowers, food, rocks, sun, tower, toy, tree, and vehicle. For image modality, 500-D bag-of-words (BOW) representation based on the SIFT feature is extracted. For text modality, we select 1000 tags with top frequency to build the dictionary. We select all the color images from *Caltech 256* [49] and choose classes having more than 80 images to form a subset *Caltech 247*. We extract two types of visual features, including Geometric Blur and DCNN feature (1024). For Geometric Blur, we use the BOW model to represent by setting the vocabulary size as 1000. DCNN feature is extracted with GoogLeNet. CUB-200-2011 [50] is represented with DCNN feature (1024) using GoogLeNet. GoogLeNet [51] is a 22-layer deep model with nine inception modules. Corel 5k [6] is represented with seven types of visual features, which fall into global ones and local ones. The former includes a color histogram (256 for gray images and 768 for color images), LBP (59), and GIST features (512), while the latter contains SIFT, dense-SIFT, Geometric Blur, and PHOG (680) descriptors. For all the local features except PHOG, we use the BOW model to represent by setting the vocabulary size as 200. The basic information of these datasets is given in Table I.

B. Experimental Setting

In the experiments, positive pairs are generated by picking a pair from the same class while negative pairs are produced by choosing a sample pair from different categories. All labeled data in the training set are used for constraints generation. The experiment is performed on a machine with a 3.40-GHz Intel processor, 8-GB memory, and MATLAB software.

C. Evaluation Criterion

We use the mean average precision (MAP) to evaluate the performance of information retrieval, which is calculated on the retrieved ranking list of queries. Given a query and its R retrieved samples, the average precision is defined as [52]

$$AP = \frac{1}{L} \sum_{r=1}^R \text{prec}(r) \delta(r)$$

where L is the number of relevant samples in the retrieved set, $\text{prec}(r)$ is the precision at the r th position, and $\delta(r)$ represents whether the r th retrieved image is relevant to the query or not. $\delta(r) = 1$ when they are relevant and 0 otherwise. The MAP is computed as the average AP of all the queries. R is set as the sample per class in the pooling set for small datasets, while it is equal to 10 for large datasets, such as *20news*, *CUB-200-2011*, *Nuswide 3k*, *Corel 5k*, and *Caltech 247*.

D. Baseline Methods

We compare our proposed SLRML and SLRMML methods with the following metric-learning algorithms as well as hashing methods.

TABLE I
DESCRIPTIONS OF DATASETS

Dataset	Type	#Class	#Feature	#Instance
Coil20 ¹	Object	20	1024	1440
text1	Text	2	7511	1946
20news ²	Text	4	8014	3970
CUB-200-2011	Image	200	1024	11788
Nuswide 3k	Image-Text	10	1500	3000
Corel 5k	Image	50	2619	5000
Caltech 247	Image	247	2024	19760

- 1) EMR is an efficient graph-based manifold ranking algorithm, which can be applied to large-scale real-world image retrieval tasks [53].
- 2) SMFH is a supervised matrix factorization hashing method which can preserve the similarities from different modalities into a produced Hamming space [54].
- 3) CCQ is a composite correlation quantization model which integrates multimodal correlation and composite quantization into a seamless latent semantic analysis framework [55].
- 4) CMM is a semisupervised distance metric-learning approach which maximizes the scatterness margin between the negative pairs and positive pairs [42].
- 5) GMML is a geometric mean metric-learning method built on geometric intuition, which has a closed-form solution [56].
- 6) DDML is a novel discriminative deep metric-learning method, which trains a deep neural network to learn a set of hierarchical nonlinear transformations [57].
- 7) OMKS is an online multiple kernel similarity learning framework that can effectively improve visual search by learning a flexible nonlinear proximity function [39].
- 8) HMML is a multiple metric-learning algorithm which jointly learns a set of homogenous/heterogeneous metrics in order to achieve fusion-based joint classification by generalizing the LMNN framework [58].
- 9) EMGMML is an efficient multimodal GMML method, which exploits both the consensus and complementarity properties among different modalities [59].
- 10) FISH-MML is a multiview metric-learning algorithm which introduces Fisher discriminant analysis to enhance separability and preserve expressiveness, and uses Hilbert–Schmidt independence criteria to promote consistency across different views [18].
- 11) MvDML is a sharable and individual multiview deep metric-learning method which utilizes the neural network to seek multiple nonlinear transformations [60].

E. Parameter Setting and Tuning

We tune the parameters on the validation sets for all the metric-learning methods. In particular, we vary the tradeoff parameter α in the range of 10^{-4} to 10^4 for both SLRML and SLRMML, and set parameter k as 10. For EMR, we set the number of the landmarks p as 50 [53]. There are several essential parameters in SMFH [54]. The length of hash

codes is set as 128 bits. The parameter λ_p controls the weight for the p th modality. In experiments, we empirically set it as $(1/m)$, where m is the number of modalities. During each iteration, the number of sampling points is set as 50 for Coil20, text1, and Nuswide 3k; 100 for 20news and Corel 5k; 200 for CUB-200-2011; and 400 for Caltech 247. The CCQ approach involves several parameters [55]. The length of hash codes is uniformly set as 64 bits. We choose to fix $\lambda = 10$ throughout the comparative study. The parameter K is set as 16 on Nuswide 3k and 256 on Caltech 247. We tune the scaling parameter β of CMM in the range of 10^{-4} to 10^4 [42]. For GMML, we set the parameter λ as 0.1. The step length t is adjusted in $[0, 1]$ with a step size 0.1. The prior matrix A_0 is set as an identity matrix [56]. For DDML, we utilize a three-layer neural network [57]. The dimension parameters of $\{W^{(m)}\}_{m=1}^M$ and $\{b^{(m)}\}_{m=1}^M$ are set as 200×300 , 150×200 , 200×1 , and 150×1 on text1 and 20news datasets, and 150×200 , 100×150 , 150×1 , and 100×1 on other datasets. There are three parameters to be tuned in OMKS, that is, the kernel parameter γ , tradeoff parameter C , and discounting parameter β . Specifically, γ is varied in $[0.01, 0.1]$ with 0.01 interval. C is adjusted in $[0.001, 0.01]$ with an incremental value 0.001 and β is tuned from 0.01 to 0.99 with an incremental value 0.01 [39]. For EMGMML, the parameter γ is tuned in the range of $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4\}$ [59]. The parameter settings of λ , A_0 , and t are the same with GMML. Both the parameters λ_1 and λ_2 of FISH-MML are tuned from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$ [18]. For MvDML, we employ a three-layer neural network ($M_k = 2$) for each view k [60], where its neural units are set as $[q_k^{(0)}, q_k^{(1)}, q_k^{(2)}] = [200, 200, 100]$ and $r^c = 50$ on Nuswide 3k; $[q_k^{(0)}, q_k^{(1)}, q_k^{(2)}] = [50, 50, 20]$ and $r^c = 10$ on Corel 5k; and $[q_k^{(0)}, q_k^{(1)}, q_k^{(2)}] = [300, 200, 150]$ and $r^c = 50$ on Caltech 247.

F. Performance Analysis

We apply our metric-learning methods to information retrieval and classification, and then compare the results with the other state-of-the-art algorithms. The parameter sensitivity analysis and time cost are also presented in the following.

1) *Information Retrieval*: For the retrieval task, we split the corresponding datasets into four subsets, including the training set (5% labeled and 45% unlabeled), validation set (10%), query set (10%), and pooling set (30%).

Table II summarizes the mAP values of different metrics for single-modal retrieval on Coil20, text1, 20news, and CUB-200-2011 datasets. SLRML-mar and SLRML-reg represent the algorithms utilizing the margin and regularizer of SLRML as objectives, respectively. It is clear that SLRML consistently outperforms all the other methods, followed by GMML and FISH-ML. On high-dimensional text datasets, such as text1 and 20news, SLRML has more significant advantages, which can be attributed to its semisupervised Laplacian regularizer. It ensures the consistency of manifold distribution and class label distribution of data, thus playing an important role in both metric learning and information retrieval tasks. SLRML-mar achieves the best performance on the CUB-200-2011 dataset.

TABLE II
mAP PERFORMANCE ON SINGLE-MODAL DATASETS. THE BEST TWO RESULTS ARE MARKED WITH RED AND BLUE, RESPECTIVELY

Metric	Coil20	text1	20news	CUB-200-2011
Euclidean	0.5183	0.3939	0.2470	0.1615
EMR	0.3175	0.2859	0.1219	0.0323
CMM	0.5388	0.2912	0.3393	0.1699
GMML	0.6139	0.6682	0.6872	0.1726
FISH-ML	0.6042	0.3936	0.7320	0.1386
DDML	0.3104	0.3228	0.7162	0.0980
SLRML-mar	0.5655	0.2995	0.2411	0.1804
SLRML-reg	0.0500	0.5000	0.2500	0.0050
SLRML	0.6212	0.7025	0.8042	0.1773

Since the number of samples is much larger than the feature dimension, the data are densely distributed in the feature space. Therefore, the effective use of discriminant information is crucial. In most cases, GMML has better performance than FISH-ML. On high-dimensional text dataset 20news, FISH-ML has better performance, which is due to its ability to learn effective low-dimensional feature space representation.

Fig. 2 shows the top- n ($n = 1, 2, \dots, 5$) precision results on Coil20 and 20news datasets. In general, SLRML is superior to all the other metric-learning algorithms, followed by FISH-ML and GMML. Unfortunately, as a semisupervised metric-learning algorithm, CMM has inferior performance with the supervised metrics. The main reason lies in that the regularizer of CMM chooses the projection direction as those with the largest variance and ignores the use of supervision information. Thus, the performance is poor in the case of less supervision information. Since larger datasets can provide more supervision information, the performance of DDML on the 20news dataset is much better than that on the Coil20 dataset.

We evaluate the retrieval performance under varied labeling rates on the text1 dataset, and then summarize the results in Fig. 3. As a semisupervised metric-learning method, SLRML achieves the best performance with a small labeling rate 0.1 and 0.2. With the labeling rate increasing, there is a significant improvement on the performance of the supervised metric-learning method GMML. When the labeling rate exceeds 0.5, GMML has a very comparable performance with SLRML. Other baseline supervised method FISH-ML and semisupervised method CMM show stable performance in the face of labeling rate changes. It is interesting to notice that the performance of DDML is obviously improved when the labeling rate reaches 0.7, which is due to the fact that the success of deep learning relies on a large number of annotated samples.

The retrieval performance on multimodal datasets is listed in Table III. Among all the methods, OMKS, HMML, EMGMML, FISH-MML, MvDML, and SLRMML are multimodal metric-learning methods which inherit the idea of subspace learning, while other approaches concatenate multiple modalities into high-dimensional feature vectors. In general, SLRMML gets the top, followed by FISH-MML and OMKS. While CMM uses unlabeled data in learning metrics,

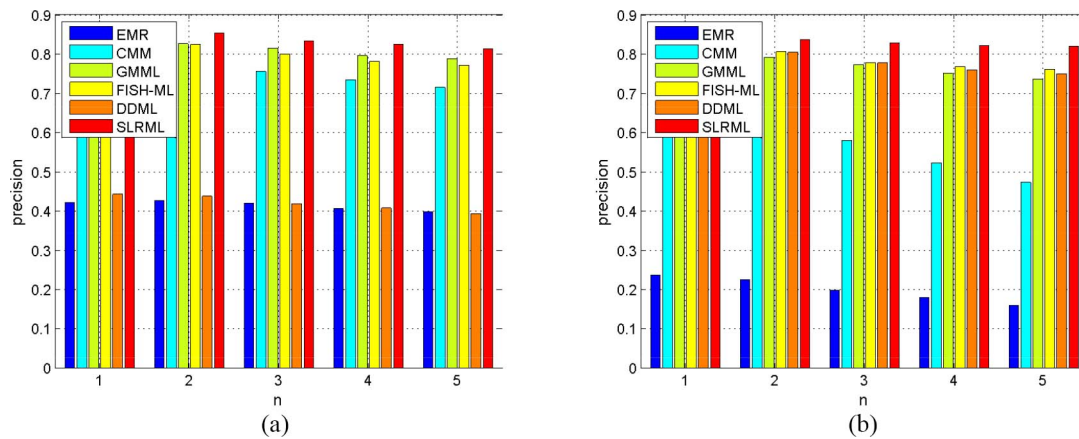


Fig. 2. Top-n precision on (a) Coil20 and (b) 20news datasets.

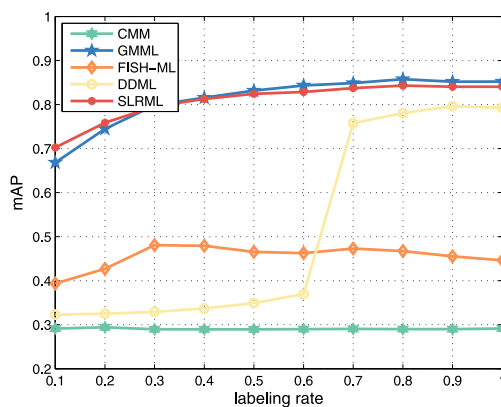


Fig. 3. mAP curves with varying labeling rates on the text1 dataset.

TABLE III
mAP PERFORMANCE ON MULTIMODAL DATASETS. THE BEST TWO RESULTS ARE MARKED WITH RED AND BLUE, RESPECTIVELY

Metric	Nuswide 3k	Corel 5k	Caltech 247
Euclidean	0.2324	0.0658	0.4130
EMR	0.0874	0.0232	0.0553
SMFH	0.4319	0.1934	0.0930
CCQ	0.2618	-	0.0187
CMM	0.2481	0.0665	0.4152
DDML	0.0575	0.0476	0.0061
OMKS	0.3974	0.0838	-
HMML	0.0673	0.0333	0.0004
EMGMML	0.4594	0.0661	0.0210
FISH-MML	0.4759	0.0705	0.2733
MvDML	0.2348	0.0433	0.0680
SLRMML	0.4899	0.0785	0.4210

the simple concatenation of feature vectors neglects the correlation between different modalities, making it fail in capturing the underlying multimodal data structure. DDML only exploits labeled data and, thus, it gets worse results. In comparison, MvDML has slightly better performance, primarily attributed to its effective use of the shared and specific properties among

multiview representations. Overall, hashing methods SMFH and CCQ do not reveal excellent performance, except in the case of Corel 5k. It is obvious that SMFH obtains the highest mAP on Corel 5k, partly due to the fact that it projects the lower-dimensional features on a higher-dimensional subspace. Moreover, the preservation of the similarities among multimodal features through a graph regularization can further boost the performance when the number of modalities increases.

We also randomly pick up a part of query images and compare the top-5 ranked images with two multimodal distance metric-learning algorithms. Fig. 4 gives the qualitative comparison results by two different methods, that is, OMKS and our SLRMML. We notice that SLRMML retrieves more relevant images than OMKS does, which implies the vital significance of exploiting unlabeled data in metric learning. Specifically, for query 1, OMKS obtains no relevant images while SLRMML retrieves four.

Fig. 5 presents the comparison results of different algorithms under varied labeling rates on the Caltech 247 dataset. As out of memory errors occur when the labeling rate increases in running DDML, OMKS, and MvDML, we remove the above algorithms from compared methods. It is clear that SLRMML consistently has a better performance compared with other metrics, followed by CMM. In contrast, supervised metric-learning methods, such as HMML, FISH-MML, and EMGMML, do not show any advantages. HMML shows unstable performance as the labeling rate increases, which is partly due to the ignorance of the importance of different modalities. For FISH-MML, its performance is also limited by the preset low dimensionality of the transformed feature subspace. While GMML obtains excellent performance in single-modal retrieval tasks, its multimodal version EMGMML fails in the learned improper combination weights. Compared with the most metric-learning algorithms, hashing methods are slightly inferior. The major reason is due to the ignorance of complementarity among different modalities, since both SMFH and CCQ focus on mapping multiple features onto a common subspace.

2) *Nearest Neighbor Classification*: For the nearest neighbor classification task, we divide the datasets into three parts,

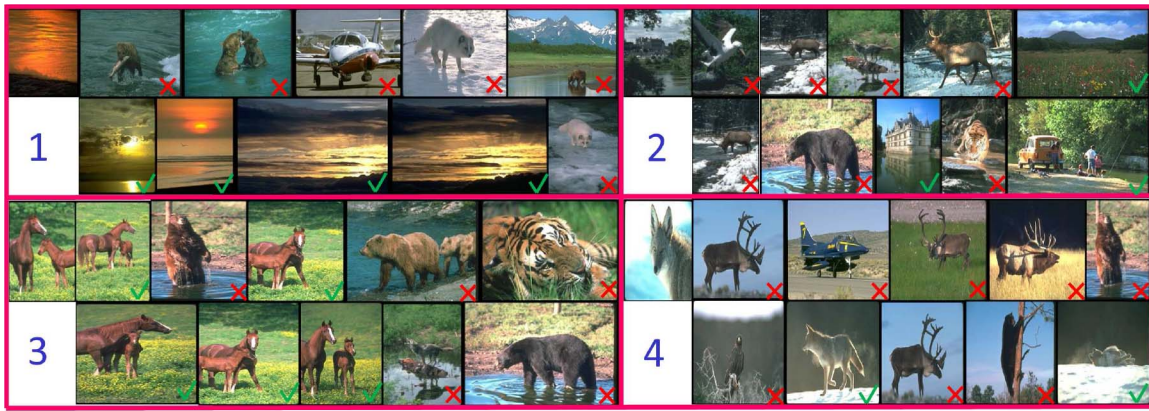


Fig. 4. Image similarity search results on the Corel 5k dataset with different metric-learning methods. Results from the first line to the second line stand for OMKS and SLRMML, respectively. “√” represents the images of the same semantic theme with the queries, and “x” represents the images from different semantic themes.

TABLE IV
CLASSIFICATION ACCURACY (%) ON SINGLE-MODAL DATASETS. THE BEST TWO RESULTS ARE MARKED WITH RED AND BLUE, RESPECTIVELY

Metric	Coil20	text1	20news	CUB-200-2011
Euclidean	69.17	79.79	33.60	24.95
CMM	72.78	85.63	55.57	28.00
GMML	71.11	90.21	73.58	28.20
FISH-ML	71.11	73.13	85.53	23.20
DDML	48.33	85.63	77.43	17.35
SLRML-mar	73.33	91.67	87.35	28.80
SLRML-reg	5.00	50.00	25.00	0.50
SLRML	71.94	91.88	87.85	28.80

TABLE V
CLASSIFICATION ACCURACY (%) ON MULTIMODAL DATASETS. THE BEST TWO RESULTS ARE MARKED WITH RED AND BLUE, RESPECTIVELY

Metric	Nuswide 3k	Corel 5k	Caltech 247
Euclidean	50.53	10.88	52.41
SMFH	28.03	4.72	0.81
CCQ	11.97	-	0.64
CMM	61.73	12.16	53.04
DDML	11.20	7.68	1.74
HMML	65.33	12.24	52.67
EMGMML	54.93	12.88	3.04
FISH-MML	60.40	10.88	38.72
MvDML	44.67	8.56	11.36
SLRMML	66.00	13.28	53.44

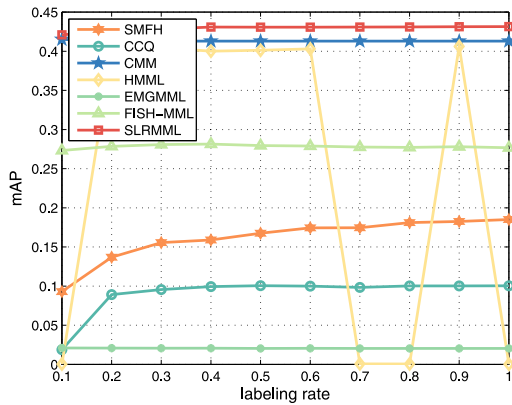


Fig. 5. mAP curves with varying labeling rates on the Caltech 247 dataset.

that is, the training set (5% labeled and 45% unlabeled), the validation set (25%), and the testing set (25%).

Classification performance of the state of the art on single-modal datasets is listed in Table IV. On the whole, SLRML is superior to other competitors, followed by GMML. Since the dimension of the reduced subspace by FISH-ML is set as 20, the ability of feature representation is weakened. Accordingly, FISH-ML is not as effective as the Euclidean metric on text1 and CUB-200-2011 datasets. We observe that the performance

of SLRML-mar and SLRML is close to each other, because the discriminative information is crucial for classification. On the contrary, SLRML-reg performs poorly. Therefore, to achieve a competitive advantage, just making use of the regularization is not enough, particularly when only a small amount of labeled data are provided.

To observe the differences of data distribution after CMM and SLRML projections, we first use CMM and SLRML to project the 20news dataset to a low-dimensional space and then give the scatter plot in Fig. 6(a) and (b), respectively. In the plot, four different colors stand for four different classes. After the projection learned with the SLRML algorithm, the interclass scatter is enlarged, particularly for the yellow ones, blue ones, and green ones, while in the CMM projected feature space, these classes are seriously overlapping.

We perform classification on multimodal datasets and report the results in Table V. The compared methods are the same as those in Table III, excluding EMR and OMKS, which are only designed for retrieval tasks. In general, SLRMML has good performance among all the competitors, followed by HMML and CMM. On the Caltech 247 dataset, the classification accuracy of EMGMML and FISH-MML is lower than that of the Euclidean distance. As EMGMML ignores

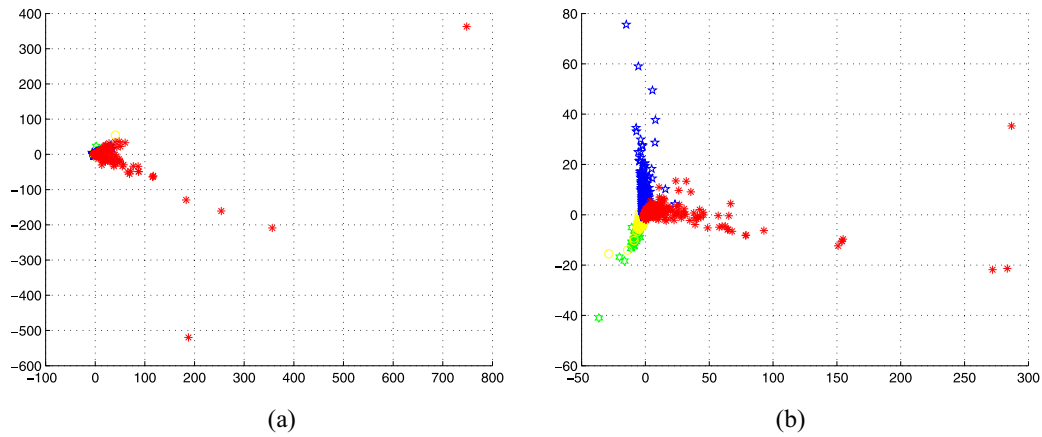


Fig. 6. Scatter plot of the 20news dataset in the (a) CMM and (b) SLRML projected space with PCA.

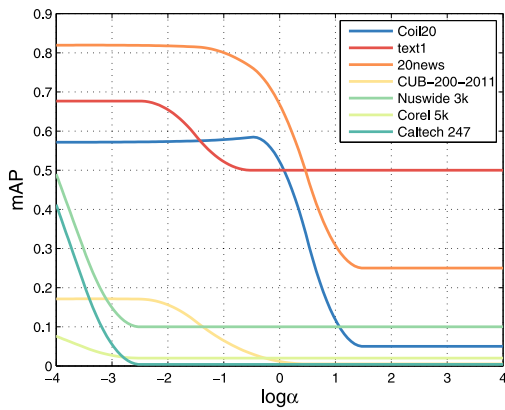


Fig. 7. mAP curves with different values of parameter α .

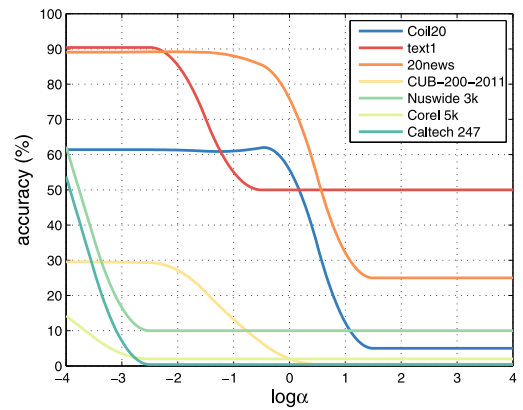


Fig. 8. Classification accuracy (%) with different values of parameter α .

the imbalance between positive and negative pairs of constraints, which affects learning performance when the number of categories is large. FISH-MML fixes the dimension of reduced feature space, resulting in severe loss of discriminant information when the number of samples is much larger than the feature dimension. The accuracies of SMFH, CCQ, and DDML are close, since they neglect the use of massive unlabeled data. In contrast, MvDML has higher accuracy, mainly due to its effective fusion strategy of multiple modalities.

3) *Parameter Analysis*: We explore the effect of parameter α on our proposed SLRML and SLRMML methods. Fig. 7 illustrates the influence of the parameter α on seven validation datasets for retrieval. It shows that good performance is obtained with a relatively smaller α . For instance, on the 20news dataset, we get the best performance by setting α as 10^{-3} , while for text1, Nuswide 3k, Corel 5k, and Caltech 247 datasets, the optimal α equals 10^{-4} .

Fig. 8 shows the influence of the parameter α on seven validation datasets for classification. Similar to Fig. 7, we notice that the best performance is obtained with smaller values of the parameter α . For instance, the optimal α equals 10^{-2} on the 20news dataset and 10^{-4} on the text1, Nuswide 3k, Corel 5k, CUB-200-2011, and Caltech 247 datasets. The experimental results indicate that the margin plays a significant role in learning appropriate metrics for retrieval and classification tasks,

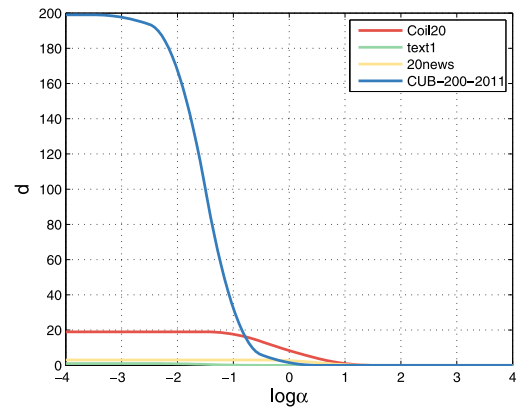


Fig. 9. Optimal dimensionality d with different values of parameter α .

whereas the regularization term seemingly has less effect on the performance.

More investigations are conducted to reveal the relation between the parameter α and the optimal dimensionality d . Fig. 9 gives the optimum d on different single-modality datasets with different values of parameter α . From the empirical results, we may conclude that when the value of parameter α increases, the optimal value of dimensionality d decreases. It is also suggested that when parameter α is fixed, a larger number of classes require higher dimensionality.

TABLE VI
RUNNING TIME (SECONDS) ON COREL 5K

SMFH	CCQ	CMM	OMKS	HMML	EMGMML	FISH-MML	SLRMML
1.71	14.96	29.63	12211.36	346.10	4.79	4.70	6.19

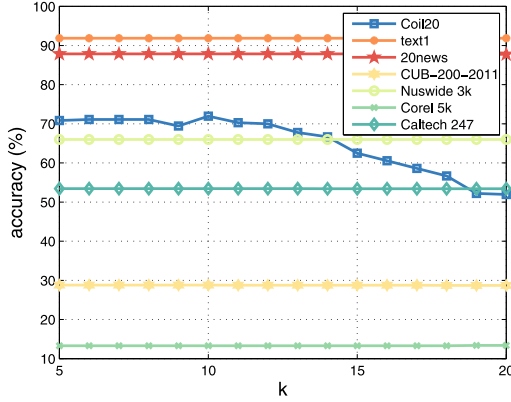


Fig. 10. Classification accuracy (%) with different values of k on seven datasets for image classification.

To further investigate the influence of parameter k on SLRML and SLRMML methods, we evaluate the classification performance of SLRML and SLRMML versus varied parameter k . Fig. 10 presents the results on seven datasets. From the plot, it is clear that the parameter k in the construction of the similarity matrix has little impact on the performance, that is, the performance of our proposed methods is not sensitive to the scaling parameter.

G. Time Complexity Analysis

In this section, we will give a brief discussion on the computational complexity of both SLRML and SLRMML when learning metrics for multimodality data.

Let D_p be the dimension of the p th modality and $D_{\max} = \max_{p \in [1, m]} D_p$. The entire length of multiple modalities is $D = \sum_{p=1}^m D_p$. The time cost of SLRML is mainly generated from two parts, that is, the computation of margin and regularizers, and the eigenvalue decomposition. Specifically, the former costs $O(DN^2)$ while the cost of the latter is $O(D^3)$. Thus, the entire time cost of SLRML is $O(D(N^2 + D^2))$. As for SLRMML, it has an additional term with respect to $\mathbf{w}$. The time of eigenvalue decomposition and computing combination weights is $O(mD_{\max}^3)$ and $O(mD_{\max}N^2)$. Therefore, the total time complexity of SLRMML is $O(mD_{\max}(D_{\max}^2 + N^2))$. Thus, SLRMML is more efficient than SLRML in handling high-dimensional multimodal tasks.

Fig. 11 plots the time costs of different metric-learning algorithms along with the increase of feature dimensions on the text1 dataset for image classification. From Fig. 10, the running time of GMML and FISH-ML grow dramatically with the dimensionality, indicating that they are not suitable for tackling high-dimensional tasks. On the contrary, CMM and our proposed SLRML are quite efficient, which can be partly due to their capability of searching the optimal dimensionality.

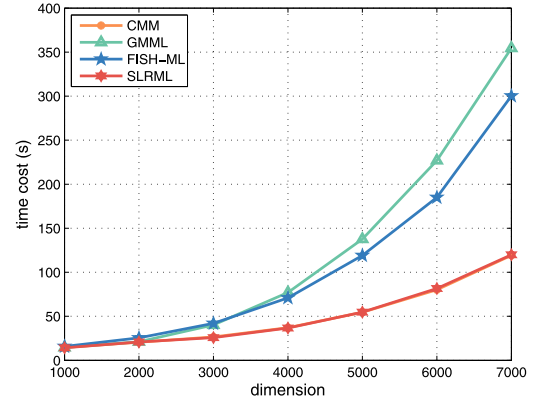


Fig. 11. Time costs (seconds) of different methods with the varying feature dimensions on the text1 dataset.

Table VI shows the running time cost of different metric-learning methods on the Corel 5k dataset for image retrieval. It should be noted that the optimal dimension of FISH-MML is set as 20 for each modality. We observe that our proposed SLRMML is considerably more scalable than most of the metric-learning methods, making it applicable for large-scale problems. In particular, SLRMML spends only less than 10 s to complete metric learning, and thus resulting in a speedup factor of more than 1000 comparisons to OMKS in terms of efficiency. The hashing methods SMFH and CCQ are also quite efficient since both of them map the original feature vectors onto a lower-dimensional subspace, that is, 128 and 64 dimensions, respectively.

VI. CONCLUSION

We have proposed an SLRML framework with limited supervision information. Our method learns metrics by minimizing the within-class scatter and maximizing the between-class scatter. Meanwhile, a semisupervised Laplace matrix is constructed to keep the consistency of the latent data manifold and the supervision information. Different from most distance metric-learning algorithms which may suffer from the curse of dimensionality, our approach utilizes the matrix decomposition technique and is suitable for solving high-dimensional problems. Empirical results show that our metric has good performance on both classification and retrieval tasks.

As there has been a flood of multimodal data recently, learning proper metrics for data represented with multiple modalities is of great significance in a lot of fields. Under such circumstances, we extend our method to deal with multiple modality data by exploiting both the complementarity and consensus among various modalities. The multimodal version first partitions the original feature space into several subspaces

according to modalities and then learns the metric in its distinctive feature subspace. Related weights are also learned from a joint optimization, representing the importance of the modality in learning metrics.

While our approach has shown advantages over some multiple kernel metric-learning methods, the linear metric may limit its performance in complex sceneries. In the future, we will consider combining deep learning to further improve the performance [61], [62]. In addition, it is also desirable to adapt our methods to multilabel and weak-label tasks.

APPENDIX PROOF OF THEOREM 3

We divide the proof problem into two subproblems, that is, 1) $\hat{U}_p$ formed with eigenvectors corresponding to the positive eigenvalues of $D_p - S_p - \alpha(X_p L_p X_p^T + I_p)$ is optimal and 2)

$$\hat{w}_p = \frac{(1/\text{tr}[U_p^T(D_p - S_p - \alpha(X_p L_p X_p^T + I_p))U_p])^{1/(r-1)}}{\sum_{p=1}^m (1/\text{tr}[U_p^T(D_p - S_p - \alpha(X_p L_p X_p^T + I_p))U_p])^{1/(r-1)}}$$

is optimal, $p = 1, 2, \dots, m$. First, because of Theorems 1 and 2, we may easily infer that $\hat{U}_p$ is optimal. Second, assume, to the contrary, $\hat{w}_p$ is not optimal. Hence, there exists $\tilde{w}_p$ such that maximizes the objective in (8). By constructing a Lagrange function $L(w, \eta)$ as (12) and setting the partial derivative in terms of w and η as 0, we obtain $r\tilde{w}_p^{r-1}\text{tr}[\hat{U}_p^T(D_p - S_p - \alpha(X_p L_p X_p^T + I_p))\hat{U}_p] - \eta = 0$ and $\sum_{p=1}^m \tilde{w}_p - 1 = 0$. Therefore, $\tilde{w}_p$ equals $\hat{w}_p$, a contradiction.

REFERENCES

- [1] N. Srivastava and R. R. Salakhutdinov, "Multimodal learning with deep Boltzmann machines," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 2222–2230.
- [2] B. McFee and G. Lanckriet, "Learning multi-modal similarity," *J. Mach. Learn. Res.*, vol. 12, pp. 491–523, Feb. 2011.
- [3] J. Liang, Q. Hu, W. Wang, and Y. Han, "Semisupervised online multi-kernel similarity learning for image retrieval," *IEEE Trans. Multimedia*, vol. 19, no. 5, pp. 1077–1089, May 2017.
- [4] Y. Jing and S. Baluja, "VisualRank: Applying PageRank to large-scale image search," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 11, pp. 1877–1890, Nov. 2008.
- [5] S. C. Hoi, W. Liu, and S.-F. Chang, "Semi-supervised distance metric learning for collaborative image retrieval," in *Proc. 21st IEEE Conf. Comput. Vis. Pattern Recognit.*, Anchorage, AK, USA, 2008, pp. 1–7.
- [6] S. C. Hoi, W. Liu, M. R. Lyu, and W.-Y. Ma, "Learning distance metrics with contextual constraints for image retrieval," in *Proc. 19th IEEE Conf. Comput. Vis. Pattern Recognit.*, New York, NY, USA, 2006, pp. 2072–2078.
- [7] A. Morales, A. Kumar, and M. A. Ferrer, "Interdigital palm region for biometric identification," *Comput. Vis. Image Understand.*, vol. 142, pp. 125–133, Jan. 2016.
- [8] J. Liang, Y. Han, and Q. Hu, "Semi-supervised image clustering with multi-modal information," *Multimedia Syst.*, vol. 22, no. 2, pp. 149–160, 2016.
- [9] N. Jiang, W. Liu, and Y. Wu, "Order determination and sparsity-regularized metric learning adaptive visual tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Providence, RI, USA, 2012, pp. 1956–1963.
- [10] C. Hu, X. Lu, M. Ye, and W. Zeng, "Singular value decomposition and local near neighbors for face recognition under varying illumination," *Pattern Recognit.*, vol. 64, pp. 60–83, Apr. 2017.
- [11] F. Nie, S. Xiang, Y. Jia, and C. Zhang, "Semi-supervised orthogonal discriminant analysis via label propagation," *Pattern Recognit.*, vol. 42, no. 11, pp. 2615–2627, 2009.
- [12] J. Liang, Q. Hu, C. Dang, and W. Zuo, "Weighted graph embedding-based metric learning for kinship verification," *IEEE Trans. Image Process.*, vol. 28, no. 3, pp. 1149–1162, Mar. 2019.
- [13] E. P. Xing, M. I. Jordan, S. J. Russell, and A. Y. Ng, "Distance metric learning with application to clustering with side-information," in *Proc. Adv. Neural Inf. Process. Syst.*, 2003, pp. 521–528.
- [14] D. M. Johnson, C. Xiong, and J. J. Corso, "Semi-supervised nonlinear distance metric learning via forests of max-margin cluster hierarchies," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 4, pp. 1035–1046, Apr. 2016.
- [15] S. Xiang, F. Nie, and C. Zhang, "Learning a mahalanobis distance metric for data clustering and classification," *Pattern Recognit.*, vol. 41, no. 12, pp. 3600–3612, 2008.
- [16] C. Hou, C. Zhang, Y. Wu, and F. Nie, "Multiple view semi-supervised dimensionality reduction," *Pattern Recognit.*, vol. 43, no. 3, pp. 720–730, 2010.
- [17] K. Q. Weinberger and L. K. Saul, "Distance metric learning for large margin nearest neighbor classification," *J. Mach. Learn. Res.*, vol. 10, pp. 207–244, Jun. 2009.
- [18] C. Zhang, Y. Liu, Y. Liu, Q. Hu, X. Liu, and P. Zhu, "Fish-MML: Fisher-HSIC multi-view metric learning," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, 2018, pp. 3054–3060.
- [19] C.-T. Do, A. Douzal-Chouakria, S. Marié, M. Rombaut, and S. Varasteh, "Multi-modal and multi-scale temporal metric learning for a robust time series nearest neighbors classification," *Inf. Sci.*, vols. 418–419, pp. 272–285, Dec. 2017.
- [20] A. Bellet, A. Habrard, and M. Sebban, "A survey on metric learning for feature vectors and structured data," Rep., 2013. [Online]. Available: arXiv:1306.6709
- [21] P. Wu, S. C. Hoi, P. Zhao, C. Miao, and Z.-Y. Liu, "Online multi-modal distance metric learning with application to image retrieval," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 2, pp. 454–467, Feb. 2016.
- [22] F. Wang and J. Sun, "Survey on distance metric learning and dimensionality reduction in data mining," *Data. Min. Knowl. Disc.*, vol. 29, no. 2, pp. 534–564, 2015.
- [23] J. Goldberger, G. E. Hinton, S. T. Roweis, and R. R. Salakhutdinov, "Neighbourhood components analysis," in *Proc. Adv. Neural Inf. Process. Syst.*, 2005, pp. 513–520.
- [24] A. Globerson and S. T. Roweis, "Metric learning by collapsing classes," in *Proc. Adv. Neural Inf. Process. Syst.*, 2006, pp. 451–458.
- [25] A. Bar-Hillel, T. Hertz, N. Shental, and D. Weinshall, "Learning a mahalanobis metric from equivalence constraints," *J. Mach. Learn. Res.*, vol. 6, pp. 937–965, Jun. 2005.
- [26] J. V. Davis, B. Kulis, P. Jain, S. Sra, and I. S. Dhillon, "Information-theoretic metric learning," in *Proc. 24th Int. Conf. Mach. Learn.*, 2007, pp. 209–216.
- [27] P. Jain, B. Kulis, I. S. Dhillon, and K. Grauman, "Online metric learning and fast similarity search," in *Proc. Adv. Neural Inf. Process. Syst.*, 2009, pp. 761–768.
- [28] R. Jin, S. Wang, and Y. Zhou, "Regularized distance metric learning: Theory and algorithm," in *Proc. Adv. Neural Inf. Process. Syst.*, 2009, pp. 862–870.
- [29] Z.-J. Zha, T. Mei, M. Wang, Z. Wang, and X.-S. Hua, "Robust distance metric learning with auxiliary knowledge," in *Proc. 21st Int. Joint Conf. Artif. Intell.*, 2009, pp. 1327–1332.
- [30] Y. Ying and P. Li, "Distance metric learning with Eigenvalue optimization," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 1–26, 2012.
- [31] K. Fukunaga, *Introduction to Statistical Pattern Recognition*. San Diego, CA, USA: Academic, 2013.
- [32] B. Liu, M. Wang, R. Hong, Z. Zha, and X.-S. Hua, "Joint learning of labels and distance metric," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 40, no. 3, pp. 973–978, Jun. 2010.
- [33] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, 2000.
- [34] I. Jolliffe, *Principal Component Analysis*. Hoboken, NJ, USA: Wiley Online Library, 2002.
- [35] J. B. Tenenbaum, V. De Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *Science*, vol. 290, no. 5500, pp. 2319–2323, 2000.
- [36] X. He and P. Niyogi, "Locality preserving projections," in *Neural Information Processing System*, vol. 16. Cambridge, MA, USA: MIT Press, 2004, pp. 153–160.
- [37] J. Lu, G. Wang, and P. Moulin, "Localized multifeature metric learning for image-set-based face recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 26, no. 3, pp. 529–540, Mar. 2016.

- [38] X. Lu, Y. Wang, X. Zhou, and Z. Ling, "A method for metric learning with multiple-kernel embedding," *Neural Process. Lett.*, vol. 43, no. 3, pp. 905–921, 2016.
- [39] H. Xia, S. C. Hoi, R. Jin, and P. Zhao, "Online multiple kernel similarity learning for visual search," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 3, pp. 536–549, Mar. 2014.
- [40] H. Tao, C. Hou, F. Nie, J. Zhu, and D. Yi, "Scalable multi-view semi-supervised classification via adaptive regression," *IEEE Trans. Image Process.*, vol. 26, no. 9, pp. 4283–4296, Sep. 2017.
- [41] W. Liu, X. Tian, D. Tao, and J. Liu, "Constrained metric learning via distance gap maximization," in *Proc. AAAI Conf. Artif. Intell.*, 2010, pp. 518–524.
- [42] F. Wang, "Semisupervised metric learning by maximizing constraint margin," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 41, no. 4, pp. 931–939, Aug. 2011.
- [43] K. Fan, "On a theorem of Weyl concerning Eigenvalues of linear transformations I," *Proc. Nat. Acad. Sci. USA*, vol. 35, no. 11, pp. 652–655, 1949.
- [44] P. Moutafis, M. Leng, and I. A. Kakadiaris, "An overview and empirical comparison of distance metric learning methods," *IEEE Trans. Cybern.*, vol. 47, no. 3, pp. 612–625, Mar. 2017.
- [45] Q. Zheng, A. Kumar, and G. Pan, "Suspecting less and doing better: New insights on palmprint identification for faster and more accurate matching," *IEEE Trans. Inf. Forensics Security*, vol. 11, no. 3, pp. 633–641, Mar. 2016.
- [46] Y. Chen and J. Su, "Sparse embedded dictionary learning on face recognition," *Pattern Recognit.*, vol. 64, pp. 51–59, Apr. 2017.
- [47] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 2012.
- [48] T.-S. Chua, J. Tang, R. Hong, H. Li, Z. Luo, and Y. Zheng, "NUS-WIDE: A real-world Web image database from national university of Singapore," in *Proc. ACM Int. Conf. Image Video Retrieval*, 2009, pp. 1–9.
- [49] G. Griffin, A. Holub, and P. Perona, "Caltech-256 object category dataset," California Inst. Technol., Pasadena, CA, USA, Rep. 7694, 2007.
- [50] C. Wah, S. Branson, P. Welinder, P. Perona, and S. Belongie, "The caltech-UCSD birds-200-2011 dataset," California Inst. Technol., Pasadena, CA, USA, Rep. CNS-TR-2011-001, 2011.
- [51] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 1–9.
- [52] Y. Zhen and D.-Y. Yeung, "A probabilistic model for multimodal hash function learning," in *Proc. 18th ACM SIGKDD Int. Conf. Knowl. Disc. Data Min.*, 2012, pp. 940–948.
- [53] B. Xu *et al.*, "Efficient manifold ranking for image retrieval," in *Proc. 34th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2011, pp. 525–534.
- [54] H. Liu, R. Ji, Y. Wu, and G. Hua, "Supervised matrix factorization for cross-modality hashing," in *Proc. 25th Int. Joint Conf. Artif. Intell.*, 2016, pp. 1767–1773.
- [55] M. Long, Y. Cao, J. Wang, and P. S. Yu, "Composite correlation quantization for efficient multimodal retrieval," in *Proc. 39th Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2016, pp. 579–588.
- [56] P. H. Zadeh, R. Hosseini, and S. Sra, "Geometric mean metric learning," in *Proc. 33rd Int. Conf. Mach. Learn.*, 2016, pp. 2464–2471.
- [57] J. Lu, J. Hu, and Y.-P. Tan, "Discriminative deep metric learning for face and kinship verification," *IEEE Trans. Image Process.*, vol. 26, no. 9, pp. 4269–4282, Sep. 2017.
- [58] H. Zhang, T. S. Huang, N. M. Nasrabadi, and Y. Zhang, "Heterogeneous multi-metric learning for multi-sensor fusion," in *Proc. 14th Int. Conf. Inf. Fusion*, 2011, pp. 1–8.
- [59] J. Liang, Q. Hu, P. Zhu, and W. Wang, "Efficient multi-modal geometric mean metric learning," *Pattern Recognit.*, vol. 75, pp. 188–198, Mar. 2018.
- [60] J. Hu, J. Lu, and Y.-P. Tan, "Sharable and individual multi-view metric learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 9, pp. 2281–2288, Sep. 2018.
- [61] B. Kulis, "Metric learning: A survey," *Found. Trends Mach. Learn.*, vol. 5, no. 4, pp. 287–364, 2013.
- [62] A. Bellet, A. Habrard, and M. Sebban, "Metric learning," *Synth. Lectures Artif. Intell. Mach. Learn.*, vol. 9, no. 1, pp. 1–151, 2015.



Jianqing Liang received the Ph.D. degree in computer application technology from Tianjin University, Tianjin, China, in 2019.

She is currently a Lecturer with the School of Computer and Information Technology, Shanxi University, Taiyuan, China. Her current research interests include metric learning, semisupervised learning, and machine learning.



Pengfei Zhu (Member, IEEE) received the B.S. and M.S. degrees from the Harbin Institute of Technology, Harbin, China, in 2009 and 2011, respectively, and the Ph.D. degree from Hong Kong Polytechnic University, Hong Kong, in 2015.

He is currently an Associate Professor with the School of Computer Science and Technology, Tianjin University, Tianjin, China. His research interests focus on machine learning and computer vision.



Chuangyin Dang (Senior Member, IEEE) received the B.S. degree in computational mathematics from Shanxi University, Taiyuan, China, in 1983, the M.S. degree in applied mathematics from Xidian University, Xi'an, China, in 1986, and the Ph.D. degree in operations research/economics from the University of Tilburg, Tilburg, The Netherlands, in 1991.

He is a Professor with the City University of Hong Kong, Hong Kong. He is best known for the development of the D1-triangulation of the Euclidean space and the simplicial method for integer programming. His current research interests include computational intelligence, optimization theory and techniques, and applied general equilibrium modeling and computation.

Prof. Dang is a member of ES, INFORMS, and MPS.



Qinghua Hu (Senior Member, IEEE) received the B.E., M.E., and Ph.D. degrees from the Harbin Institute of Technology, Harbin, China, in 1999, 2002, and 2008, respectively.

He once worked with the Harbin Institute of Technology as an Assistant Professor and an Associate Professor from 2006 to 2011 and a Postdoctoral Fellow with Hong Kong Polytechnic University, Hong Kong. He is currently a Full Professor with Tianjin University, Tianjin, China. He has published more than 100 journal and conference papers in the areas of pattern recognition, machine learning, and data mining. His research interests are focused on intelligent modeling, data mining, and knowledge discovery for classification and regression.

Prof. Hu is the PC Co-Chair of RSCTC 2010, CRSSC 2012, and ICMLC 2014 and serves as a referee for many journals and conferences.